

Lyapunov-Controlled Minimax Policy Learning for Adversarial Dynamic Spectrum Access

Jialin Zhuang and Vincent K. N. Lau

Department of Electronic and Computer Engineering, The Hong Kong University of Science and Technology
Hong Kong SAR, China; Emails: jzhuangag@connect.ust.hk; eeknlu@ece.ust.hk

Abstract—Adaptive jamming makes dynamic spectrum access (DSA) adversarial because the transmitter and jammer adapt their channel-selection policies. We model it as a zero-sum Markov game and propose Lyapunov-Controlled Minimax Policy Learning (LCMPL) with separate neural policies. LCMPL combines the simultaneous minimax policy-gradient field with a Jacobian–field correction, decouples their learning rates, and jointly regulates them using local game geometry and a communication-aware merit. A curvature identity identifies when the correction reduces the saddle-field energy that measures joint policy stationarity. Under local smoothness, biased-oracle moment bounds, and geometric coverage by the field and curvature directions, the time-averaged expected stationarity residual has an $O(1/K)$ transient plus an explicit residual term, where K is the number of joint transmitter–jammer updates. Under a local policy-gap condition, a communication corollary converts this learning-stability result into a bound on the transmitter’s worst-case spectral-efficiency loss. Across the nominal and trajectory-sampled benchmarks, LCMPL attains the highest final mean worst-case spectral-efficiency return among the compared methods. With sampled saddle fields, it improves the final rate over its field-only ablation by 0.0223 bit/s/Hz, with a paired 95% confidence interval of [0.0138, 0.0308].

Index Terms—Anti-jamming communications, dynamic spectrum access, reinforcement learning, minimax policy learning, zero-sum Markov games.

I. INTRODUCTION

Dynamic spectrum access (DSA) lets a secondary transmitter select a channel based on incumbent activity and channel conditions [1], while accounting for fading, interference, switching overhead, and adaptive jamming [2], [3], [4].

Because each learned policy changes the opponent’s future data and gradient, adversarial DSA combines exogenous radio variation, endogenous opponent adaptation, and stochastic oracle uncertainty. The central problem is to stabilize joint policy adaptation without sacrificing spectral efficiency.

A two-player zero-sum Markov game captures this interaction [5], [6]. The transmitter policy π_θ , with parameter vector θ , maximizes a discounted spectral-efficiency return, while the jammer policy ν_ψ , with parameter vector ψ , minimizes it through $\max_\theta \min_\psi J(\theta, \psi)$. Their policy gradients form a saddle field; worst-case transmitter return measures communication robustness, while exploitability measures sensitivity to further policy adaptation.

This work was supported by the Research Grants Council (RGC), Hong Kong, under the Areas of Excellence Scheme Grant AoE/E-101/23-N and AoE/E-601/22-R. (Corresponding author: Vincent K. N. Lau.)

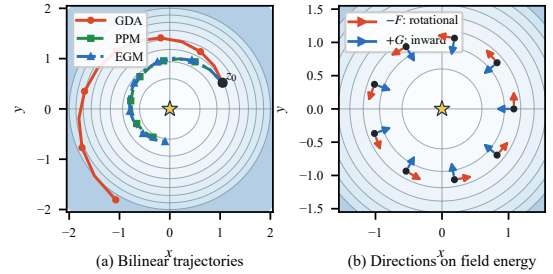


Fig. 1. Geometry of curvature stabilization. (a) Simultaneous gradient descent–ascent (GDA) rotates in a bilinear game, whereas the proximal point method (PPM) and extragradient method (EGM) contract. (b) The negative saddle field is tangent to field-energy contours, while its Jacobian–field correction points inward; this separation motivates independent control of the two directions.

As Fig. 1 illustrates, simultaneous gradient descent–ascent (GDA) can cycle on a skew-dominant joint field [7]. The proximal point method (PPM) evaluates the field implicitly, whereas the extragradient method (EGM) uses an extrapolated pair [8], [9]. Their local expansions include a Jacobian–field action but couple the effective field and curvature scales through one learning rate [10]. Adaptive Mirror-Prox methods use observed operator geometry to tune extragradient updates across smoothness regimes [11], while double-step extragradient assigns distinct scales to the extrapolation and update stages [12]. Our formulation instead treats the policy-gradient field and its Jacobian–field correction as separate search coordinates whose local coupling changes with the wireless state and neural policies.

Motivated by adaptive learning-rate and Lyapunov-drift control [13], [14], we use the predicted one-step change of a communication-aware Lyapunov merit to coordinate geometric stabilization, worst-case-rate loss, and exploitability.

In this paper, we develop Lyapunov-Controlled Minimax Policy Learning (LCMPL) for adaptive anti-jamming DSA. LCMPL assigns separate rates to the joint policy-gradient field and its Jacobian–field correction by minimizing a local model of this change. A box-constrained quadratic model proposes both rates, and a same-state safeguard retains curvature only when it does not worsen realized merit or worst-case rate relative to the field-only candidate. We prove an $O(1/K)$ stationarity transient with an explicit residual and, under a local policy-gap condition, bound worst-case-rate loss; four regimes test the learning rules.

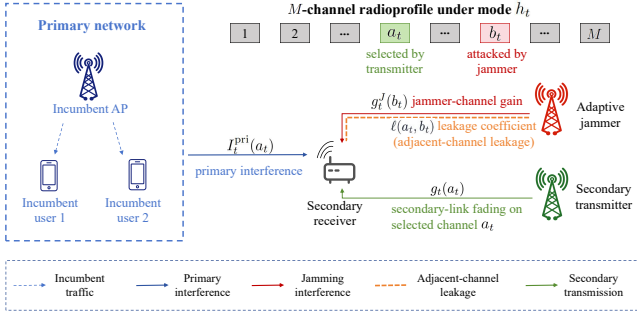


Fig. 2. Adversarial wireless environment for anti-jamming dynamic spectrum access.

II. ADVERSARIAL DSA SYSTEM AND MARKOV-GAME MODEL

A. Multi-Channel Wireless Environment and Stage Utility

Fig. 2 shows an incumbent network, a secondary link, and an adaptive jammer over M licensed subchannels; $h_t \in \mathcal{H} = \{1, \dots, L\}$ indexes one of L compressed radio modes. Actions a_t, b_t select channels from $\mathcal{A} = \mathcal{B} = \{0, \dots, M-1\}$. Write the channel-indexed gain and interference profiles as $g_t = \{g_t(a)\}_{a \in \mathcal{A}}$ and I_t , where I_t contains incumbent interference $I_t^{\text{pri}}(a)$ and jammer-receiver gains $g_t^J(b)$. In the general model, g_t and I_t may depend on h_t , and the observed state is $s_t = (h_t, g_t, I_t, a_{t-1}) \in \mathcal{S}$, where a_{t-1} determines switching cost. In the reported specialization, I_t is fixed by a normalized background and unit jammer-receiver gain, while h_t indexes only $g(h_t)$. The distinct gain profile and one-hot a_{t-1} give both neural policies a one-to-one encoding of the LM compact states; both actions are selected simultaneously from s_t .

The constants P_s, P_J , and N_0 denote secondary transmission power, jammer interference level, and receiver noise power, respectively. The factor $\ell(a, b)$ models co-channel, adjacent-channel, and nonadjacent leakage, and $c_{\text{sw}} \geq 0$ is the switching cost. For actions (a_t, b_t) , the received signal-to-interference-plus-noise ratio (SINR) and switching-cost-adjusted stage spectral efficiency are

$$\text{SINR}_t = \frac{P_s g_t(a_t)}{N_0 + I_t^{\text{pri}}(a_t) + P_J g_t^J(b_t) \ell(a_t, b_t)}, \quad (1)$$

$$r_t = \log_2(1 + \text{SINR}_t) - c_{\text{sw}} \mathbf{1}_{\{a_t \neq a_{t-1}\}}.$$

B. Radio Dynamics and Minimax Policy Objective

The radio mode follows an action-independent correlated Markov chain with transition matrix P_h , and the compact benchmark transition is

$$P((h', a') | (h, a_{\text{prev}}, a, b) = P_h(h'|h) \mathbf{1}_{\{a'=a\}}.$$

We use $P_h(h|h) = \rho_h$ and $P_h(h'|h) = (1 - \rho_h)/(L - 1)$ for $h' \neq h$, where ρ_h controls temporal correlation. The reported finite benchmarks set $L = M$. The initial state is uniform over the LM compact states. This specialization makes the exogenous chain control correlated secondary-link fading, while the adaptive jammer creates endogenous interference through its selected channel and leakage profile. Let $\pi_\theta(a|s)$ and $\nu_\psi(b|s)$

denote differentiable transmitter and jammer policies. Their normalized discounted return is

$$J(\theta, \psi) = (1 - \delta) \mathbb{E} \left[\sum_{t=0}^{\infty} \delta^t r_t \right],$$

where $\delta \in (0, 1)$ is the discount factor. To preserve stochastic channel exploration during joint adaptation, we train both policies on the entropy-regularized return:

$$J_\tau = J + (1 - \delta) \tau \mathbb{E} \left[\sum_{t=0}^{\infty} \delta^t \{H(\pi_\theta(\cdot|s_t)) - H(\nu_\psi(\cdot|s_t))\} \right],$$

where $\tau > 0$ is the entropy weight and $H(\cdot)$ is the Shannon entropy. Thus all learning updates and the Lyapunov merit use J_τ , whereas the separate rate safeguard and reported worst-case communication performance use J . Let Π_T and Π_J denote the stationary Markov policy spaces of the transmitter and jammer, respectively. The learned policies are neural, while evaluation best responses range over these full stationary-policy spaces. The primary communication metric is the transmitter's worst-case normalized discounted spectral-efficiency return,

$$R_{\text{BR}}(\theta) = \min_{\nu \in \Pi_J} J(\pi_\theta, \nu).$$

The reported unregularized policy exploitability is

$$\Delta(z) = \max_{\pi \in \Pi_T} J(\pi, \nu_\psi) - \min_{\nu \in \Pi_J} J(\pi_\theta, \nu).$$

Replacing J by J_τ defines the regularized gap Δ_τ used in the merit. Dynamic programming computes R_{BR} and Δ , while soft Bellman recursion computes Δ_τ ; the remaining problem is stable simultaneous adaptation.

III. GEOMETRY-AWARE MINIMAX POLICY LEARNING

A. Curvature-Induced Two-Direction Learning

For the joint parameter $z = (\theta, \psi)$, define the saddle field

$$\mathbf{F}(z) = \begin{bmatrix} -\nabla_\theta J_\tau(\theta, \psi) \\ \nabla_\psi J_\tau(\theta, \psi) \end{bmatrix}. \quad (2)$$

For a learning rate $s > 0$, $z_+ = z - s\mathbf{F}(z)$ ascends in the transmitter policy and descends in the jammer policy, while either update changes the opponent's gradient. We assess GDA through the field energy $\phi(z) = \|\mathbf{F}(z)\|^2/2$, a scaled squared stationarity residual. For GDA, $\phi(z - s\mathbf{F}) = \phi(z) - s\langle \nabla \phi, \mathbf{F} \rangle + O(s^2)$. The first-order decrease can vanish when the transmitter-jammer Jacobian is dominated by its antisymmetric component, allowing GDA to rotate while $\mathbf{F}$ remains nonzero. PPM and EGM respond to this field variation by evaluating the field after a policy displacement:

$$\begin{aligned} \text{GDA: } z_+^{\text{GDA}} &= z - s\mathbf{F}(z), \\ \text{PPM: } z_+^{\text{PPM}} &= z - s\mathbf{F}(z_+^{\text{PPM}}), \\ \text{EGM: } \tilde{z} &= z - s\mathbf{F}(z), \quad z_+^{\text{EGM}} = z - s\mathbf{F}(\tilde{z}), \end{aligned}$$

Here, $\tilde{z}$ is EGM's extrapolation point; PPM uses an implicit next-policy gradient, while EGM uses an explicit predictor. Let $\nabla \mathbf{F}(z)$ denote the Jacobian of the saddle field. Expanding the

displaced field in EGM, or the implicit field in PPM, gives $z_+ = z - s\mathbf{F}(z) + s^2\nabla\mathbf{F}(z)\mathbf{F}(z) + O(s^3)$. Their common correction motivates the Jacobian-field direction

$$\mathbf{G}(z) = \nabla\mathbf{F}(z)\mathbf{F}(z). \quad (3)$$

Thus, $\mathbf{G}$ predicts how simultaneous transmitter-jammer movement changes the next joint policy gradient; Section IV identifies when it counters rotational coupling. Their second-order expansions couple the field and curvature terms through the fixed relation (s, s^2) . Because the geometry can change independently along these directions, LCMPL assigns field and curvature learning rates β_k and γ_k :

$$z_{k+1} = z_k - \beta_k \hat{\mathbf{F}}_k + \gamma_k \hat{\mathbf{G}}_k. \quad (4)$$

Here, $\hat{\mathbf{F}}_k$ and $\hat{\mathbf{G}}_k$ are stochastic oracles for $\mathbf{F}(z_k)$ and $\mathbf{G}(z_k)$, respectively. In the trajectory-sampled realization, the same batch defines the empirical field and the Jacobian-vector product (JVP)

$$\hat{\mathbf{G}}_k = \nabla\hat{\mathbf{F}}_k(z_k)\hat{\mathbf{F}}_k(z_k).$$

The biased-oracle model in Section IV accounts for the resulting same-batch JVP error. The curvature direction introduces no additional wireless action; it modifies only the policy update using local game geometry. The remaining design problem is to choose β_k and γ_k without trading policy stabilization for communication performance.

B. Communication-Aware Lyapunov Policy Update

Because field energy measures stationarity rather than communication quality, LCMPL evaluates both directions through a Lyapunov merit combining field energy, exploitability, and worst-case-rate deficit. Define the normalized communication loss

$$\mathcal{L}_c(z) = \lambda_\Delta \left(\frac{\Delta_\tau(z)}{\Delta_{\tau,0}} \right)^2 + \lambda_R \left(\frac{D_{R,\tau}(\theta)}{D_0} \right)^2,$$

where $\lambda_\Delta, \lambda_R \geq 0$ are merit weights and $\Delta_{\tau,0}, D_0 > 0$ are initial scales. Let $r_{\max}$ be the largest stage reward in the finite game and define the entropy-regularized stage-return ceiling $R_{\text{ub},\tau} = r_{\max} + \tau \log M$. The deficit $D_{R,\tau} = R_{\text{ub},\tau} - \min_{\nu \in \Pi_j} J_\tau(\pi_\theta, \nu) \geq 0$ uses this ceiling for normalization. The Lyapunov merit is

$$V(z) = \frac{\phi(z)}{\phi_0} + \mathcal{L}_c(z), \quad (5)$$

where $\phi_0 > 0$ is the initial field-energy scale. Its predicted one-step change supplies a common criterion for geometric stabilization and communication quality.

Let $\mathbf{d}_{k,1} = -\hat{\mathbf{F}}_k$, $\mathbf{d}_{k,2} = \hat{\mathbf{G}}_k$, and $\mathbf{u} = (\beta, \gamma)^\top$. For positive probe radii $q_{k,1}$ and $q_{k,2}$, define $V_{k,i}^\pm = V(z_k \pm q_{k,i} \mathbf{d}_{k,i})$ and $V_k^0 = V(z_k)$. Centered finite differences set $[\mathbf{a}_k]_i = (V_{k,i}^+ - V_{k,i}^-)/(2q_{k,i})$ and

$$[\mathbf{H}_k]_{ii} = \max \left\{ \frac{V_{k,i}^+ - 2V_k^0 + V_{k,i}^-}{q_{k,i}^2}, h_{\min} \right\}$$

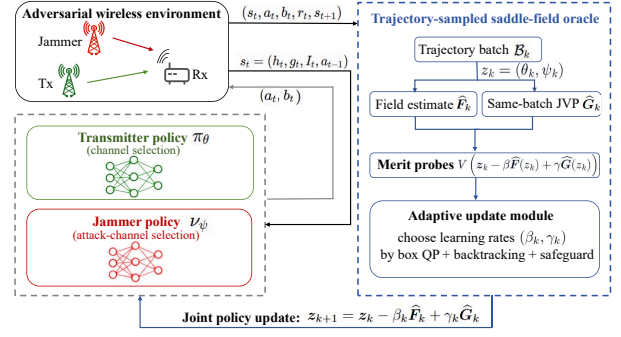


Fig. 3. LCMPL minimax policy-learning framework.

for $i \in \{1, 2\}$, with $[\mathbf{H}_k]_{12} = [\mathbf{H}_k]_{21} = 0$ and $h_{\min} > 0$. The diagonal model avoids mixed-direction probes. If $\hat{\mathbf{G}}_k = 0$, set $[\mathbf{a}_k]_2 = 0$ and $[\mathbf{H}_k]_{22} = h_{\min}$; the QP then reduces to the field-only model. Let $\bar{\beta} > 0$ and $0 < \gamma_{\min} \leq \gamma_{\max}$ be deterministic global rate bounds, and let the oracle-dependent cap $\gamma_{\min} \leq \bar{\gamma}_k \leq \gamma_{\max}$ be measurable after the current oracle query. LCMPL selects its learning-rate pair with the box-constrained quadratic program (QP)

$$m_k(\mathbf{u}) = \mathbf{a}_k^\top \mathbf{u} + \frac{1}{2} \mathbf{u}^\top \mathbf{H}_k \mathbf{u}, \quad \tilde{\mathbf{u}}_k = \arg \min_{\mathbf{u} \in \mathcal{U}_k} m_k(\mathbf{u}), \quad \mathcal{U}_k = [0, \bar{\beta}] \times [0, \bar{\gamma}_k]. \quad (6)$$

Let $\mathbf{u}_k^F$ minimize m_k over $\mathcal{U}_F = [0, \bar{\beta}] \times \{0\}$. Define the extra fitted-model decrease supplied by the curvature coordinate as $D_{G,k}^{\text{box}} = m_k(\mathbf{u}_k^F) - m_k(\tilde{\mathbf{u}}_k) \geq 0$. LCMPL independently backtracks $\tilde{\mathbf{u}}_k$ and $\mathbf{u}_k^F$ from the same z_k until the realized merit does not exceed $V(z_k)$; a failed search returns the zero step. The resulting pairs are $\bar{\mathbf{u}}_k$ and $\bar{\mathbf{u}}_k^F$. The safeguard sets $\mathbf{u}_k^{\text{acc}} = \bar{\mathbf{u}}_k$ when the curvature candidate has no larger realized V and no smaller R_{BR} ; otherwise, $\mathbf{u}_k^{\text{acc}} = \bar{\mathbf{u}}_k^F$. Set $(\beta_k, \gamma_k)^\top = \mathbf{u}_k^{\text{acc}}$ in Eq. (4). The LCMPL-F ablation keeps the merit, field-direction QP and probes, β bound, and backtracking rule but omits $\hat{\mathbf{G}}_k$ and fixes $\gamma_k = 0$. Fig. 3 summarizes the loop in which communication feedback forms a trajectory batch $\mathcal{B}_k$ for the field and Jacobian-action estimates.

IV. LEARNING STABILITY AND COMMUNICATION GUARANTEES

The analysis first determines when $+\mathbf{G}$ decreases field energy, then connects the accepted decrease in the local Lyapunov model to finite-time communication performance. Write $\nabla\mathbf{F} = \mathbf{S} + \mathbf{W}$, where $\mathbf{S} = (\nabla\mathbf{F} + \nabla\mathbf{F}^\top)/2$ and $\mathbf{W} = (\nabla\mathbf{F} - \nabla\mathbf{F}^\top)/2$.

Proposition 1 (Curvature identity for adversarial policy learning): At every point where $\mathbf{F}$ is differentiable, the Jacobian-field direction satisfies

$$\langle \nabla\phi, \mathbf{G} \rangle = \|\mathbf{S}\mathbf{F}\|^2 - \|\mathbf{W}\mathbf{F}\|^2. \quad (7)$$

Hence, $+\mathbf{G}$ decreases the field energy when $\|\mathbf{W}\mathbf{F}\| > \|\mathbf{S}\mathbf{F}\|$.

Proof: With $\mathbf{A} = \nabla\mathbf{F}$, $\langle \nabla\phi, \mathbf{G} \rangle = \mathbf{F}^\top \mathbf{A}^2 \mathbf{F}$. Expanding $\mathbf{A} = \mathbf{S} + \mathbf{W}$ eliminates the cross term because $\mathbf{S}\mathbf{W} + \mathbf{W}\mathbf{S}$ is antisymmetric. Moreover, $\mathbf{F}^\top \mathbf{S}^2 \mathbf{F} = \|\mathbf{S}\mathbf{F}\|^2$ and $\mathbf{F}^\top \mathbf{W}^2 \mathbf{F} = -\|\mathbf{W}\mathbf{F}\|^2$, which gives Eq. (7). $\square$

Proposition 1 gives a deterministic geometric condition for curvature stabilization, whereas LCMPL uses stochastic field estimates and finite-difference merit probes. We therefore separate estimation errors from geometric Lyapunov conditions. Let $\mathcal{F}_k$ denote the history available immediately before drawing the stochastic-oracle and probe samples at update k . The analysis starts from an explicit biased-oracle model.

Assumption 1 (Local regularity and stochastic oracles):

- (a) A compact set $\mathcal{T}$ contains all accepted iterates, trial segments, and probe points, and $\mathbf{F} \in C^1$ and $V \in C^3$ near $\mathcal{T}$, with $\|\nabla V\| \leq G_V$, $\|\nabla^2 V\| \leq H_V$, and $\|\nabla^3 V\| \leq M_3$.
- (b) The stochastic oracles satisfy $\hat{\mathbf{F}}_k = \mathbf{F}(z_k) + \mathbf{b}_k^F + \boldsymbol{\zeta}_k^F$ and $\hat{\mathbf{G}}_k = \mathbf{G}(z_k) + \mathbf{b}_k^G + \boldsymbol{\zeta}_k^G$, where the biases are $\mathcal{F}_k$ -measurable and $\mathbb{E}[\boldsymbol{\zeta}_k^F | \mathcal{F}_k] = \mathbb{E}[\boldsymbol{\zeta}_k^G | \mathcal{F}_k] = 0$.
- (c) For $X \in \{F, G\}$, finite nonnegative $\mathcal{F}_k$ -measurable bounds $b_{X,k}$, $\sigma_{2,k}$, and $\sigma_{3,k}$ satisfy $\|\mathbf{b}_k^X\| \leq b_{X,k}$, $\mathbb{E}[\|\boldsymbol{\zeta}_k^X\|^2 | \mathcal{F}_k] \leq \sigma_{2,k}^2$, and $\mathbb{E}[\|\boldsymbol{\zeta}_k^X\|^3 | \mathcal{F}_k] \leq \sigma_{3,k}^3$.

The same-batch JVP error is allowed in $\mathbf{b}_k^G$ and is not assumed unbiased.

Define the population field-energy decrease signals

$$\chi_F(\mathbf{z}) = \mathbf{F}^\top \mathbf{S} \mathbf{F}, \quad \chi_G(\mathbf{z}) = -\langle \nabla \phi(\mathbf{z}), \mathbf{G}(\mathbf{z}) \rangle.$$

Positive χ_F and χ_G certify decrease of ϕ along $-\mathbf{F}$ and $+\mathbf{G}$, respectively.

Assumption 2 (Geometric coverage): The local coverage modulus

$$m_{\text{geo}} = \inf_{\substack{\mathbf{z} \in \mathcal{T} \\ \|\mathbf{F}(\mathbf{z})\| > 0}} \frac{\max\{\chi_F(\mathbf{z}), \chi_G(\mathbf{z})\}}{\|\mathbf{F}(\mathbf{z})\|^2} \quad (8)$$

satisfies $m_{\text{geo}} > 0$.

Let $\sigma_{\min}$ denote the smallest singular value. At a target stationary point $\mathbf{z}^*$, the conditions $\mathbf{S}(\mathbf{z}^*) \succeq 0$ and $\sigma_{\min}(\nabla \mathbf{F}(\mathbf{z}^*)) > 0$ imply Eq. (8) on a sufficiently small neighborhood by compactness of the unit sphere and continuity; this includes nonsingular pure rotation, where $\chi_F = 0$ and $\chi_G > 0$. Define $D_F = \langle \nabla V, \mathbf{F} \rangle$ and $D_G = -\langle \nabla V, \mathbf{G} \rangle$. Compactness and smoothness make

$$B_c = \sup_{\mathbf{z} \in \mathcal{T}} \max\{|\langle \nabla \mathcal{L}_c, \mathbf{F} \rangle|, |\langle \nabla \mathcal{L}_c, \mathbf{G} \rangle|\} < \infty.$$

Because $V = \phi/\phi_0 + \mathcal{L}_c$, selecting the larger energy signal gives

$$\max\{D_F(\mathbf{z}), D_G(\mathbf{z})\} \geq \mu_{\text{geo}} \|\mathbf{F}(\mathbf{z})\|^2 - B_c, \quad (9)$$

$$\mu_{\text{geo}} = m_{\text{geo}}/\phi_0.$$

To state the drift bound, let $s \in (0, \min\{\bar{\beta}, \gamma_{\min}\}]$ be a deterministic pure-coordinate comparator and let $q_{k,i} \leq \bar{q}_i$. For $\mathbf{d}_{k,1} = -\hat{\mathbf{F}}_k$ and $\mathbf{d}_{k,2} = \hat{\mathbf{G}}_k$, the C^3 condition bounds the centered first- and second-coefficient errors by $M_3 q_{k,i}^2 \|\mathbf{d}_{k,i}\|^3/6$ and $M_3 q_{k,i} \|\mathbf{d}_{k,i}\|^3/3$. For $X \in \{F, G\}$, let $\hat{\mathbf{X}}_k$ denote the corresponding oracle, and for $p \in \{2, 3\}$ set $M_{p,X,k} = \mathbb{E}[\|\hat{\mathbf{X}}_k\|^p | \mathcal{F}_k]$. Set $b_{D,k} = \max\{b_{F,k}, b_{G,k}\}$ and $M_{p,D,k} = \max\{M_{p,F,k}, M_{p,G,k}\}$. The omitted mixed Hessian term is bounded by $R_k^{\text{cross}} = H_V \beta_k \gamma_k \|\hat{\mathbf{F}}_k\| \|\hat{\mathbf{G}}_k\|$. Define

the fitted-model loss from backtracking or safeguarding as $R_k^{\text{acc}} = m_k(\mathbf{u}_k^{\text{acc}}) - m_k(\tilde{\mathbf{u}}_k) \geq 0$. The residual below groups communication-loss variation, oracle bias and noise, probe error, the omitted mixed Hessian term, and safeguard or backtracking loss. With $(X_1, X_2) = (\mathbf{F}, \mathbf{G})$, $(u_1^{\text{max}}, u_2^{\text{max}}) = (\bar{\beta}, \gamma_{\text{max}})$, $C_{2,k} = \sqrt{M_{2,F,k} M_{2,G,k}}$, and $A_k = \mathbb{E}[R_k^{\text{acc}} | \mathcal{F}_k]$, the conditional probe, Taylor, and mixed-term bounds yield the $\mathcal{F}_k$ -measurable residual

$$\begin{aligned} \mathcal{R}_k(s) = & s(G_V b_{D,k} + B_c) + \frac{1}{2} s^2 (H_V M_{2,D,k} + h_{\min}) \\ & + M_3 \sum_{i=1}^2 [C_{\text{pr}}(\bar{q}_i^2 + \bar{q}_i) + \frac{2}{3} (u_i^{\text{max}})^3] M_{3,X_i,k} \\ & + H_V \bar{\beta} \gamma_{\text{max}} C_{2,k} + A_k, \end{aligned} \quad (10)$$

Here, $C_{\text{pr}} < \infty$ is a deterministic finite-difference constant determined by the learning-rate box. Thus, $\mathcal{R}_k$ is $\mathcal{F}_k$ -measurable and bounds all random residuals after conditioning.

Proposition 2 (Conditional Lyapunov drift): Under Assumptions 1–2, the implemented accepted update satisfies

$$\begin{aligned} \mathbb{E}[V(\mathbf{z}_{k+1}) - V(\mathbf{z}_k) | \mathcal{F}_k] \\ \leq -s \mu_{\text{geo}} \|\mathbf{F}(\mathbf{z}_k)\|^2 + \mathcal{R}_k(s). \end{aligned}$$

Proof: At update k , select the larger of $D_F(\mathbf{z}_k)$ and $D_G(\mathbf{z}_k)$ only for the analysis. Because both pure-coordinate pairs $(s, 0)$ and $(0, s)$ are box feasible, the fitted QP decrease dominates the selected comparator. Taylor expansion adds the probe, cubic, safeguard, and omitted mixed-Hessian residuals bounded in Eq. (10). Conditional expectation removes the zero-mean errors, bounds the remaining oracle terms by their moments, and Eq. (9) supplies the population decrease. $\square$

Theorem 1 (Main finite-time minimax stationarity result): Under Assumptions 1–2, every integer $K \geq 1$ satisfies

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \|\mathbf{F}(\mathbf{z}_k)\|^2 \leq \frac{V(\mathbf{z}_0)}{s \mu_{\text{geo}} K} + \frac{\bar{\mathcal{R}}_K(s)}{s \mu_{\text{geo}}}, \quad (11)$$

where $\bar{\mathcal{R}}_K = K^{-1} \sum_{k=0}^{K-1} \mathbb{E} \mathcal{R}_k$.

Proof: Take total expectations in Proposition 2, sum over k , and use $V(\mathbf{z}_K) \geq 0$. $\square$

If $\sup_k \mathbb{E} \mathcal{R}_k(s) \leq \bar{R} < \infty$, Theorem 1 gives an $O(1/K)$ transient plus the floor $\bar{R}/(s \mu_{\text{geo}})$. The result covers pure rotation, where the field coordinate has zero first-order effect and the curvature coordinate supplies the strict decrease predicted by Fig. 1 and Proposition 1. Here, s is an analytical comparator, not an implemented learning rate.

The following local compatibility condition bridges Theorem 1 from neural parameter-space stationarity to full-policy best responses.

Let $v^* = \max_{\pi \in \Pi_T} \min_{\nu \in \Pi_J} J(\pi, \nu)$ denote the unregularized stationary-policy game value. Since each M -ary policy has entropy at most $\log M$, the regularized and unregularized returns differ by at most $\tau \log M$ for either player. Consequently,

$$0 \leq v^* - R_{\text{BR}}(\boldsymbol{\theta}) \leq \Delta_\tau(\mathbf{z}) + 2\tau \log M. \quad (12)$$

Let B_K denote the right-hand side of Eq. (11).

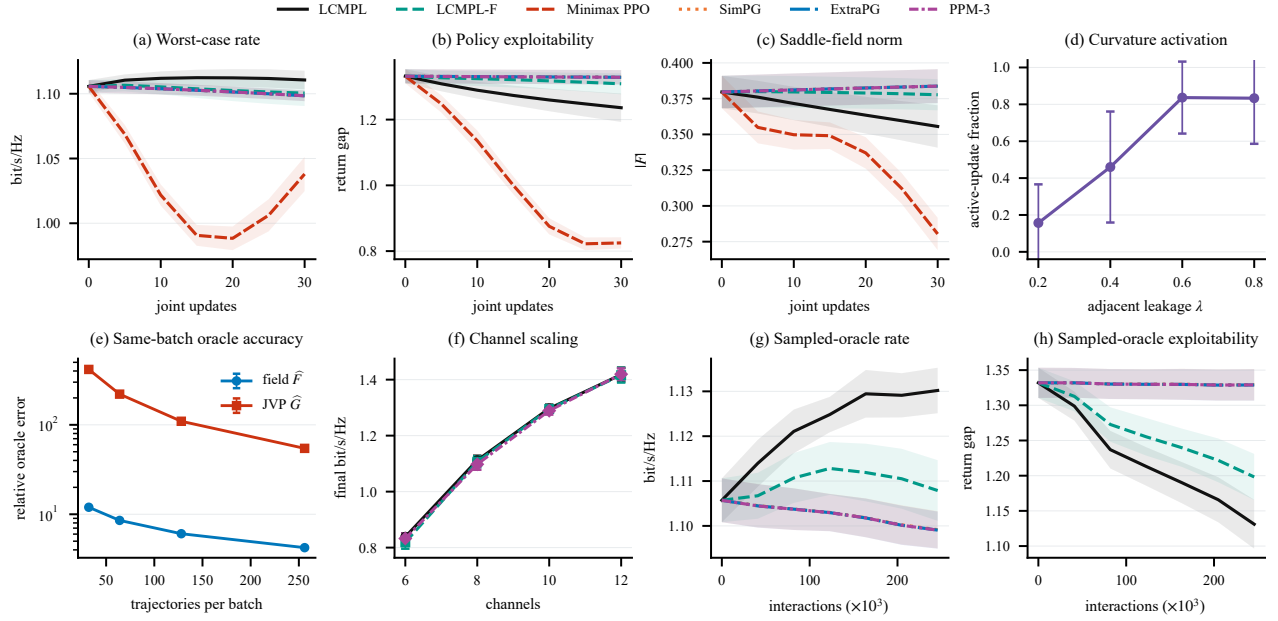


Fig. 4. Adversarial DSA results: (a)–(c) nominal population dynamics; (d) leakage-dependent curvature activation; (e) oracle fidelity; (f) channel scaling; and (g)–(h) equal-interaction trajectory learning. Means use 12 seeds except eight in (f); bands in (a)–(c) and (g)–(h) show one standard error, and bars in (d)–(f) show 95% confidence intervals.

Corollary 1 (Main communication result: worst-case spectral efficiency): If nonnegative constants C_{gap} and ξ_{gap} satisfy $\Delta_{\tau}(z) \leq C_{\text{gap}} \|F(z)\|^2 + \xi_{\text{gap}}$ on $\mathcal{T}$, then

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}[v^* - R_{\text{BR}}(\theta_k)] \leq C_{\text{gap}} B_K + \xi_{\text{gap}} + 2\tau \log M.$$

Proof: Average Eq. (12), apply the local gap condition, and invoke Theorem 1. $\square$

Corollary 1 bounds worst-case spectral-efficiency loss under local compatibility between neural stationarity and the policy-space gap; the last term is the entropy-regularization error.

V. COMMUNICATION PERFORMANCE EVALUATION

We use worst-case normalized discounted spectral-efficiency return as the primary metric and unregularized exploitability Δ , field norm, and oracle error as diagnostics.

A. Adversarial DSA Scenario and Evaluation Protocol

Wireless scenario: We evaluate the link and adaptive jammer in Fig. 2. The nominal benchmark uses $M = L = 8$, 12 fixed seeds, $\rho_h = 0.75$, $N_0 = 1$, $I_t^{\text{pri}}(a) = 0$, $g_t^J(b) = 1$, $(P_s, P_J, c_{\text{sw}}) = (10, 20, 0.10)$, $\delta = 0.95$, and $\tau = 0.02$. Its base secondary-link gain vector is $g_m^{\text{base}} = 1.9(0.18/1.9)^{m/(M-1)}$, $m = 0, \dots, M-1$, and the $L = M$ radio modes are cyclic shifts of this vector. The jammer-to-noise ratio is $\rho_J = P_J/N_0$ and equals 20 nominally; adjacent-channel leakage $\lambda = 0.80$ enters the cyclic profile $1, \lambda, 0.12, 0.025$ at channel distances $0, 1, 2, >2$. The channel-scale study uses $M = L \in \{6, 8, 10, 12\}$, while the 12-seed wireless-stress study uses $M = L = 6$ and varies ρ_J , c_{sw} , λ , and ρ_h one factor at a time.

Policies and comparison protocol: Both players use independent two-layer tanh–softmax policies with the fully observed state encoding in Section II. The nominal setting uses 64 hidden units, 1608 parameters per player, and 30 joint updates; stress tests use 32 units and 25 updates. The nominal benchmark compares simultaneous policy gradient (SimPG), extragradient policy gradient (ExtraPG), a three-step PPM approximation (PPM-3), minimax proximal policy optimization (Minimax PPO), LCMPL-F, and LCMPL. They share the environment, policies, initialization, test seeds, entropy coefficient, outer-update count, and simultaneous start without warm-up; the five saddle-field rules also share the entropy-regularized policy-gradient oracle. Minimax PPO is a population clipped-surrogate reference with root-mean-square-normalized exact advantages and three clip-0.2 epochs of fresh per-update Adam [15]. Disjoint validation seeds select rates 0.01 for the fixed-step rules and 10^{-3} for Minimax PPO from prespecified grids. LCMPL-F is the field-only ablation defined in Section III-B.

LCMPL configuration: Let $f_k = \|\hat{F}_k\|$ and $g_k = \|\hat{G}_k\|$. We use $q_{k,1} = \min\{0.012, 0.02/\max(f_k, 10^{-12})\}$ and clip $q_{k,2} = q_{k,1}f_k/g_k$ to $[2 \times 10^{-4}, 0.025]$. We set $\bar{\beta} = 0.08$, $\gamma_{\min} = 5 \times 10^{-4}$, $\gamma_{\max} = 0.12$, $\bar{\gamma}_k = \text{clip}(0.08f_k/g_k, \gamma_{\min}, \gamma_{\max})$, and $h_{\min} = 10^{-6}$; at most eight backtracking contractions by 1/2 seek a merit change no larger than 10^{-10} . The merit uses weights $(\lambda_{\Delta}, \lambda_R) = (0.35, 1.5)$, and its components are normalized by their positive initial values.

B. Nominal Anti-Jamming Performance

LCMPL’s final mean worst-case rate is 1.1106 bit/s/Hz, versus 1.1005 for LCMPL-F, 1.0984 for PPM-3, and 1.0380 for Minimax PPO in Fig. 4(a). Its paired gains over LCMPL-F

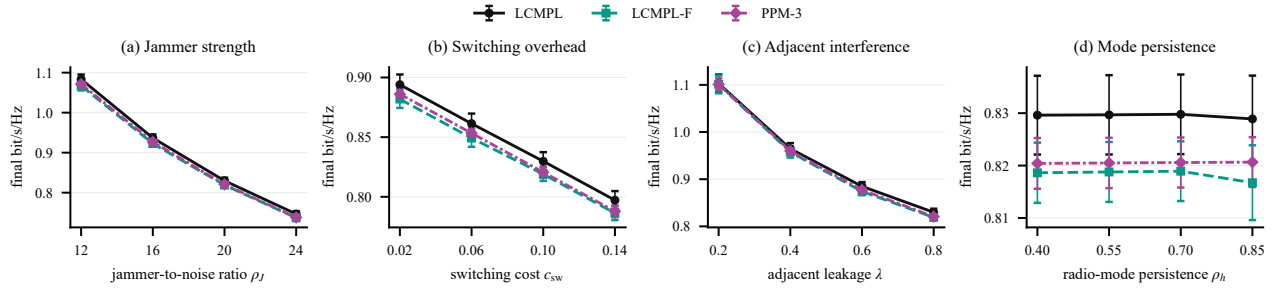


Fig. 5. Wireless stress across jammer strength, switching cost, leakage, and mode persistence: 12-seed means and 95% confidence intervals.

are 0.0102 bit/s/Hz in rate with a 95% confidence interval of $[0.0006, 0.0197]$ and 0.0728 in exploitability reduction with interval $[0.0346, 0.1109]$. Minimax PPO reaches smaller final optimization residuals but a lower rate, showing that stationarity alone does not ensure communication robustness.

C. Channel Scalability and Stochastic-Oracle Fidelity

Fig. 4(f) shows that LCMPL exceeds LCMPL-F in final mean worst-case rate across 6–12 channels. Panel (e) compares sampled and exact finite-game oracles at updates 0, 15, and 30 over 12 seeds using $e_X = \|\hat{X} - X\| / \max\{\|X\|, 10^{-12}\}$ for $X \in \{F, G\}$. Increasing the batch from 32 to 256 trajectories reduces mean relative field and JVP errors by 64.7% and 86.9%, respectively.

D. Trajectory-Sampled Anti-Jamming Learning

Panels (g) and (h) test the trajectory-sampled realization in Fig. 3. Each update uses 128 length-64 on-policy trajectories, and every method receives 245,760 transitions per seed. Both players share one batch; automatic differentiation forms $\hat{G}_k$, while ExtraPG and PPM-3 reuse the batch through per-decision likelihood ratios. Exact finite-game dynamic programming evaluates the merit, safeguard, and evaluation checkpoints. At the final budget, LCMPL improves worst-case rate over LCMPL-F by 0.0223 bit/s/Hz with a paired 95% confidence interval of $[0.0138, 0.0308]$ and reduces exploitability by 0.0672 with interval $[0.0440, 0.0903]$. LCMPL also attains a higher rate and lower exploitability than SimPG, ExtraPG, and PPM-3. After safeguarding, $\gamma_k > 10^{-10}$ on 25.3% of 360 updates, while the field-only candidate is selected on another 24.2%.

E. Robustness Across Wireless Operating Points

Fig. 4(d) shows 12-seed leakage-dependent curvature activation, while Fig. 5 retrains LCMPL, LCMPL-F, and PPM-3 under one-factor changes in jammer-to-noise ratio, switching cost, leakage, and mode persistence. LCMPL has the highest final mean rate in every plotted sweep cell, with paired mean gains of 0.0019–0.0161 bit/s/Hz over LCMPL-F.

VI. CONCLUSION

LCMPL separates field and curvature adaptation in adversarial DSA. In rotational regimes, curvature supplies Lyapunov decrease, while the same-state safeguard retains it only when

it does not underperform the field-only candidate. The biased-oracle analysis bounds time-averaged stationarity and, under a local policy-gap condition, worst-case spectral-efficiency loss; LCMPL achieves the highest final mean worst-case rate across the reported regimes.

REFERENCES

- [1] Q. Zhao, B. Krishnamachari, and K. Liu, “On myopic sensing for multi-channel opportunistic access: Structure, optimality, and performance,” *IEEE Trans. Wireless Commun.*, vol. 7, no. 12, pp. 5431–5440, Dec. 2008.
- [2] O. Naparstek and K. Cohen, “Deep multi-user reinforcement learning for distributed dynamic spectrum access,” *IEEE Trans. Wireless Commun.*, vol. 18, no. 1, pp. 310–323, Jan. 2019.
- [3] X. Liu, Y. Xu, L. Jia, Q. Wu, and A. Anpalagan, “Anti-jamming communications using spectrum waterfall: A deep reinforcement learning approach,” *IEEE Commun. Lett.*, vol. 22, no. 5, pp. 998–1001, May 2018.
- [4] X. Wang, J. Wang, Y. Xu, J. Chen, L. Jia, X. Liu, and Y. Yang, “Dynamic spectrum anti-jamming communications: Challenges and opportunities,” *IEEE Commun. Mag.*, vol. 58, no. 2, pp. 79–85, Feb. 2020.
- [5] L. S. Shapley, “Stochastic games,” *Proc. Natl. Acad. Sci. U.S.A.*, vol. 39, no. 10, pp. 1095–1100, Oct. 1953.
- [6] M. L. Littman, “Markov games as a framework for multi-agent reinforcement learning,” in *Proc. 11th Int. Conf. Mach. Learn. (ICML)*, Jul. 1994, pp. 157–163.
- [7] D. Balduzzi, S. Racaniere, J. Martens, J. Foerster, K. Tuyls, and T. Graepel, “The mechanics of n-player differentiable games,” in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, vol. 80, Jul. 2018, pp. 354–363.
- [8] R. T. Rockafellar, “Monotone operators and the proximal point algorithm,” *SIAM J. Control Optim.*, vol. 14, no. 5, pp. 877–898, Aug. 1976.
- [9] G. M. Korpelevich, “The extragradient method for finding saddle points and for other problems,” *Ekonomika i Matematicheskie Metody*, vol. 12, no. 4, pp. 747–756, 1976.
- [10] H. Lu, “An $O(s^r)$ -resolution ODE framework for understanding discrete-time algorithms and applications to the linear convergence of minimax problems,” *Math. Program.*, vol. 194, pp. 1061–1112, 2022.
- [11] K. Antonakopoulos, E. V. Belmega, and P. Mertikopoulos, “Adaptive extra-gradient methods for min-max optimization and games,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Virtual Conf., May 2021.
- [12] Y.-G. Hsieh, F. Iutzeler, J. Malick, and P. Mertikopoulos, “Explore aggressively, update conservatively: Stochastic extragradient methods with variable stepsize scaling,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, Virtual Conf., Dec. 2020, pp. 16 223–16 234.
- [13] J. Zhang, C. Jin, and Y. Gu, “Adaptive Polyak step-size for momentum accelerated stochastic gradient descent with general convergence guarantee,” *IEEE Trans. Signal Process.*, vol. 73, pp. 462–476, Jan. 2025.
- [14] Y. Yuan, D. H. K. Tsang, and V. K. N. Lau, “Step size adaptation for accelerated stochastic momentum algorithm using SDE modeling and Lyapunov drift minimization,” *IEEE Trans. Signal Process.*, vol. 73, pp. 3124–3139, Jul. 2025.
- [15] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” 2017, [Online]. Available: <https://arxiv.org/abs/1707.06347>.